

**Bosonic quantum error correction using cat qubits.** The building block of AWS' Ocelot architecture is the cat qubit, a technology inspired by the famous Schrodinger's cat thought equation. Cat qubits store quantum states using oscillators, which generate a repetitive electrical signal with steady timing.

In classical systems, the state of an oscillator is represented by the amplitude and phase of its oscillations. In quantum mechanics, the amplitude and phase are never simultaneously perfectly defined, and there is an underlying graininess to the amplitude associated with quanta of energy known as bosonic particles.

The best-known bosonic particle is the photon, associated with electromagnetic fields. We can access more oscillator states (amplitude) by pumping more energy into the system and creating more bosons (photons). Bosonic quantum error correction uses these extra oscillator states to protect quantum information and correct errors more effectively and efficiently.

Ocelot implements bosonic quantum error correction with cat qubits. As explained by AWS scientists in *Amazon Science*, "Cat qubits use the quantum superposition of classical-like states of well-defined amplitude and phase to encode a qubit's worth of information." The rate of bit-flip errors in cat qubits can be made exponentially small by increasing the number of protons in the oscillator. So, instead of increasing the qubit count, we just need to increase the energy in the oscillator to make error correction more efficient.

In the Ocelot architecture, bit-flip errors are significantly suppressed at the physical qubit level itself. Phase-flip errors are corrected using repetition codes, one of the simplest classical error-correcting codes. Additionally, highly noise-biased

controlled-NOT (CNOT) gates are placed between each cat qubit and ancillary transmon qubits (the conventional qubit used in superconducting quantum circuits) to enable phase-flip error detection while preserving the bit-flip protection.

**Promising architecture.** Ocelot demonstrates exceptional performance for superconducting qubits, with bit-flip times nearing one second and phase-flip times of 20 $\mu$ s.

It consists of two integrated silicon microchips, each measuring roughly 1cm<sup>2</sup>. They are bonded one on top of the other in an electrically-connected chip stack. Thin layers of superconducting materials form the quantum circuit elements on the surface of each silicon microchip. The chip consists of five cat qubits (data qubits), five buffer circuits to stabilise the data qubits, and four additional qubits to detect errors in the data qubits. The oscillators, which are critical to the bosonic quantum error correction implemented in Ocelot, are made using a tantalum-based superconducting material, and standard chip industry techniques are used to fabricate the chip. This is a big breakthrough, which makes the architecture sustainable and scalable.

The team claims that scaling Ocelot could reduce the quantum error correction overhead by nearly 90% compared to traditional surface code methods with similar physical-qubit error rates. Also, the tailor-made noise-biased repetition code used in Ocelot can significantly reduce the number of physical qubits required. It requires only nine physical qubits to form one logical qubit, which brings down the hardware requirements. The team believes that their approach could yield useful quantum computers with only 100,000 qubits rather than a million.

"In the future, quantum chips built according to the Ocelot

## Why do we need quantum computing?

The purpose of a quantum computer is not simply to solve classical computing problems faster than a supercomputer—although it can. Quantum computers excel in solving problems that require analysing a very large number of possible outcomes, including ones that are considered intractable in classical computing—like the Travelling Salesman problem. Solving such problems using quantum computers could significantly impact areas like:

- Optimisation
- Cryptography and cyber security
- Artificial intelligence and machine learning
- Sensing
- Telecommunications
- Credit scoring and investment risk analysis
- Simulation of natural processes like protein folding, molecular formations, photosynthesis, and superconductivity
- Drug discovery and clinical trial acceleration
- Disease risk prediction and mitigation
- Weather modelling and seismic imaging
- Structural design for buildings
- Material sciences and chemical engineering

architecture could cost as little as one-fifth of current approaches, due to the drastically reduced number of resources required for error correction. Concretely, we believe this will accelerate our timeline to a practical quantum computer by up to five years," said Oskar Painter, director of quantum hardware at AWS, in an Amazon report, while also admitting that they are just getting started and have several more stages of scaling to go through.

We can see that most incumbents are only in the early stages of their roadmaps to developing fault-tolerant, large quantum computers for commercial applications. There is still a long distance to cover. Still, the journey is an important one—because everyone is convinced that quantum computers are uniquely poised to unravel some of Nature's greatest mysteries and help solve some of the greatest challenges facing our generation, from climate change to food shortage. **EFY**